

Wandi Xu

(She/Her/Hers)

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EDUCATION

Johns Hopkins University

Ph.D. in Chemistry (Computational Chemistry)

M.A. in Chemistry

Baltimore, MD

Expected May 2027

May 2025

University of Minnesota, Twin Cities

B.S. in Chemistry GPA: 3.89/4.00

Minneapolis, MN

January 2022

RESEARCH EXPERIENCE

Johns Hopkins University - Hernandez Group

Baltimore, MD

Graduate Researcher

August 2022 – Present

- Built multiscale molecular-dynamics models of lipid vesicles (all-atom, MARTINI, DPD) and validated their structural and dynamical consistency across scales (manuscript in revision, *RSC Advances*).
- Simulated MPA-coated gold nanoparticles binding proteins to identify binding orientations and key interfacial residues (manuscript in preparation).
- Modeled MUA-/MUTAB-functionalized gold nanoparticles to characterize ligand-island formation on mixed-ligand surfaces.
- Ran periodic DFT on Ce-substituted metal-organic frameworks to quantify dopant effects for a collaborator.

University of Minnesota - Institute on the Biology of Aging & Metabolism

Minneapolis, MN

Research Assistant Advisors: Prof. Paul Robbins, Prof. Lei Zhang

October 2020 – January 2022

- Ran cell-culture, transfection, and drug-treatment workflows for non-homologous end-joining (NHEJ) assays.
- Extracted RNA from mouse tissues and ran qPCR to evaluate a candidate anti-aging compound.
- Optimized assay conditions and quantified SIRT6 activity for a self-mono-ADP-ribosylation study.

TECHNICAL SKILLS

Computational Chemistry

Molecular dynamics (GROMACS, LAMMPS, NAMD); all-atom & coarse-grained MD (MARTINI, DPD); DFT & quantum chemistry (Quantum ESPRESSO, ORCA)

Programming & Computing

Python (NumPy, Pandas, SciPy, MDAnalysis), MATLAB; machine learning (PyTorch); Git, Bash, $\text{\LaTeX}$, HPC / SLURM

Data Analysis & Visualization

VMD, ChimeraX, OVITO, VESTA, Blender; ChemDraw, MestReNova, ImageJ

Laboratory

Cell culture, transfection, PCR / qPCR, gel electrophoresis, NanoDrop, microscopy

Languages

English (fluent), Mandarin Chinese (native)

LEADERSHIP EXPERIENCE

Center for Sustainable Nanotechnology

Executive Committee Liaison

June 2024 – June 2025

- Elected by students and postdocs to represent them in Executive Committee and administrative meetings with the leadership of this NSF-funded nanotechnology research center.
- Organized and moderated recurring student and postdoc board meetings and served as the liaison between trainees and Center management.

Johns Hopkins University

Baltimore, MD

Teaching Assistant, *Physical Chemistry I & II*

Fall 2025 – Spring 2026

- Led problem-solving sessions and held office hours for a section of ~ 27 students across two upper-level physical chemistry courses.

Teaching Assistant, *General Chemistry*

Fall 2022 – Fall 2024

- Supported large-enrollment general chemistry across five semesters, leading review sessions, grading exams, and holding weekly office hours.

New Oriental Vision Overseas Consulting Co. Ltd.

Beijing, China

Teaching Assistant — SAT Preparation

August 2019 – September 2019

- Taught SAT preparation to a group class of high-school students.

Private Tutoring

Qinhuangdao, China

Instructor — Mathematics, Physics, Chemistry

October 2017 – August 2018

- Developed lesson plans and provided one-on-one tutoring in mathematics, physics, and chemistry to junior-high students.

ACADEMIC SERVICE

- Peer reviewer, *Scientific Reports* (3 manuscripts).

PUBLICATIONS

W. Xu, *et al.* "[Title:Structural and Dynamical Consistency in Lipid Vesicles across All-Atom, MARTINI, and Dissipative Particle Dynamics]." *RSC Advances* (in revision, 2026).

W. Xu, *et al.* "[Title: binding interactions between MPA-coated gold nanoparticles and proteins]." In preparation (2026).